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Dr. Seifert
Klinik und Poliklinik für Viszeral-, Thorax- und Gefäßchirurgie
Universitätsklinikum Carl Gustav Carus Dresden
Ref. 1776

Re: Wissenschaftliche*r Mitarbeiter*in — Tumor Immunology / Gastric Cancer (Ref. 1776)

Dear Dr. Seifert,

I am applying for the research associate position in your group (Ref. 1776), studying immune metabolism of tumor-infiltrating T cells in gastric cancer. My interest in this position is specific: the computational tools I have built for immunopeptidology — neoantigen prediction, MHC binding, and immune escape analysis — are directly applicable to the T cell biology questions the group addresses, and they are tools that experimental tumor immunology groups rarely have in-house. My biochemistry and pharmaceutical biotechnology training provides the lab skills foundation; the computational capability for T cell epitope analysis and immune escape modelling is what I bring that is unusual for this role.

Computational tools for T cell immunology in gastric cancer. The central problem in understanding T cell responses to gastric tumors is identifying which neoantigens drive productive anti-tumor T cell recognition, which tumor-derived variants allow the tumor to escape T cell killing, and which TCR-pMHC interactions are strong enough to sustain effector function in the metabolically hostile tumor microenvironment. I have built and deployed all three of these analyses as live tools:

The neoantigen predictor identifies which somatic mutations produce peptides that are simultaneously presented on MHC (spectral DFT cosine similarity $\rho_{\mathrm{MHC}} \geq 0.72$) and immunogenically distinct from self-peptides ($\Delta\rho_{\mathrm{self}} = \rho(\mathrm{WT}, \mathrm{self}) - \rho(\mathrm{mut}, \mathrm{self}) > 0.15$). For gastric cancer, this provides a rapid computational screen of the tumor mutational burden to rank neoantigens by their likelihood of driving productive $\mathrm{CD8}^{+}$ T cell responses — before running in vitro binding assays or organoid co-culture experiments.

The immune escape predictor systematically screens all single amino acid substitutions in a target epitope for variants that retain receptor binding while reducing antibody or T cell recognition. The escape score $E(v) = -\max \rho(v, \mathrm{Ab}) + w \cdot \rho(v, \mathrm{rec})$ ($w = 0.80$) identifies substitutions that simultaneously escape immune pressure and maintain functional interaction. In the gastric cancer context, this is the computational equivalent of what the organoid system provides experimentally: a model for why immune-edited tumor clones survive T cell surveillance.

The MHC binding predictor operates at $\mathcal{O}(cL \log L)$ complexity, allowing rapid scanning of patient-specific HLA alleles against the full neoantigen landscape of a tumor. No database lookup is required — binding is determined from spectral complementarity between peptide and HLA physicochemical channels. Combined with flow cytometry data from tumor-infiltrating T cell subpopulation analysis, this provides a closed loop: compute predicted binders → validate presentation by pMHC tetramer staining → correlate with T cell exhaustion markers (PD-1, TIM-3, LAG-3) in the patient organoid system.

Immune metabolism and T cell exhaustion context. The T cell metabolic reprogramming in the tumor microenvironment — the switch from oxidative phosphorylation to glycolysis, and the downstream effects on T cell effector function and exhaustion — has a natural representation in the disease state framework I have developed. The Syndrome framework models cellular state as $\mathbf{D} \in [0, 1]^8$ across eight oscillator dimensions including ATP synthesis coherence, genetic expression state, and circadian regulatory coherence — three dimensions directly relevant to T cell metabolic phenotype. A trajectory on this manifold represents the progression from effector to exhausted T cell state under metabolic pressure; the sparse ℓ_1 LP identifies the minimum intervention (metabolic rescue, checkpoint blockade, or combination) required to redirect the trajectory. This framework provides a quantitative language for connecting the metabolic measurements the group makes (oxygen consumption rate, glycolytic flux, mitochondrial membrane potential) to T cell functional outcomes.

Cancer biology and biochemistry background. My BSc in Biochemistry and Cell Biology (Jacobs University Bremen) and MSc in Pharmaceutical Biotechnology (HAW Hamburg) together cover the practical biochemistry and cell biology toolkit: cell culture, protein analysis, assay development, and immunological techniques from training. At the University Medical Center Groningen (2013) I performed UPLC-MS/MS shotgun proteomics for cancer biomarker discovery — work that required developing and validating sample preparation protocols for tumour-derived material. In my doctoral research at TUM I worked on mass spectrometry-based molecular profiling at scale, including federated processing of clinical multi-omics datasets. The Lavoisier pipeline I subsequently built handles spectral proteomics and metabolomics data at >100 GB scale (98.9 % NIST accuracy), applicable to the metabolomic profiling of T cell metabolic state that the position's research programme involves.

Contribution to the group. Experimental tumor immunology groups generate large and structurally complex datasets — flow cytometry, scRNA-seq, TCR repertoire sequencing, proteomics, and organoid imaging. The computational layer for integrating these modalities, identifying immunogenic neoantigens, predicting escape variants, and modelling T cell state trajectories is typically not available in-house in a surgical oncology research group. The tools I bring address exactly these bottlenecks, and they are available immediately, deployed, and validated. I would contribute both at the bench (cell culture, flow cytometry data acquisition and analysis, histological staining and quantification) and at the analysis layer where the computational tools create leverage across all experimental outputs.

I hold a BSc in Biochemistry and Cell Biology, an MSc in Pharmaceutical Biotechnology, and an MSc in Computational Biology (Universität Tübingen). I am legally resident in Germany with full work authorisation. English is native; German is conversational. I am available immediately and willing to relocate to Dresden.

Yours sincerely,

Kundai Farai Sachikonye